Would you rather?
be able to...
speak every language
or speak with animals?

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Q1
~Code~
340

CS 340

REST API and X as a Service

Updates

1. Check your grades on Canvas

1. MP 8 - Weather Filter, due next week

1. HW 8 and 9 due next Wednesday

REST API and X as a Service + passwords

Today's LGs - Build up your mental model of the websites and APIs

Be able to identify if an API is following REST principles
Be able to describe the three X as a Service types
Be able to take steps to host your own website
Be able to name one consideration of cloud computing at scale
Be able to describe the basics of how to store a password

Agenda

1. REST APIs

1. X as a service

1. Distributed Computing

1. Storing Passwords

When you use a web API your program is the (XXXX) that creates and sends an HTTP request.

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A. Client

B. Server

The request is split into packets and routed across the internet to the correct computer based on the XXXX.

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Q3

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A. TCP header

B. UDP header

C. IP address

D. HTTP information

The packets arrive at the correct computer and are sent to the correct port which is found in the packet's XXXX layer.

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Q4

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A. Internet Layer (IP)

B. Transport Layer (TCP)

C. Payload (HTTP)

The API is a (XXXX). It is listening at a specific port and receives the request, processes the information, and responds with a HTTP response.

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Q5

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A. Client

B. Server

Web API

application programming interface

server + docs

REST - Representational state transfer

set of principles or conventions that help you work with an API

REST Principles

HTTP requests and responses

Uniform Interface

simplicity and predictability

Stateless

-scalability



REST Uniform Interface - HTTP Requests

Methods are GET, PUT, POST, DELETE

↓ verbs

URL is the resource ← endpoint specifies the resource

Bad ← not Restful

GET url/getUser?id=123

GET url/users/123

REST Uniform Interface - HTTP Requests

```python
@routes.get('/getUser')
async def get_user(request):
    user_id = request.rel_url.query['id']
    return web.json_response({"id": user_id, "name": "Alice"})
```

Bad

```python
@routes.get('/users/{id}')
async def get_user(request):
    user_id = request.match_info['id']
    return web.json_response({"id": user_id, "name": "Alice"})
```

good

REST Uniform Interface - HTTP Responses

2XX - Success

4XX - client error - 404

5XX - Server error

Usually return JSON or nothing

REST - Stateless

Everything the server need is in a single request

Why? – Can spread traffic across multiple servers
- if one server it's fine.

Stateless Misnomer -

Can store state in an external database

Is this RESTful?

```
class Game:
    def __init__(self):
        self.word = "flower"
        self.lives = 5
        self.guesses = []
        self.status = ["-"] * 6

    def guess(self, letter):
        # logic of what to do with a guess
        return None

grm = Game()
@routes.get("/")
async def get_page(request: web.Request) -> web.Response:
    return web.Response(text=open('spring_game.html').read(), content_type='text/html; charset=utf-8')

@routes.post("/guess")
async def post_guess(request: web.Request) -> web.Response:
    letter = await request.text()
    result = grm.guess(letter)
    return web.Response(text=result)
```

no

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```python
@routes.post("/game")
async def create_game(request: web.Request) -> web.Response:
    game = {"word": "flower", "lives": 5, "guesses": [], "status": ["-"] * 6}
    game_id = save_game(game)  # save to database
    return web.json_response({"game_id": game_id}, status=201)

@routes.get("/game/{game_id}")
async def get_game(request: web.Request) -> web.Response:
    game_id = request.match_info['game_id']
    game = load_game(game_id)  # fetch from database
    return web.json_response(game)

@routes.post("/game/{game_id}/guess")
async def post_guess(request: web.Request) -> web.Response:
    game_id = request.match_info['game_id']
    game = load_game(game_id)  # fetch from database
    body = await request.json()
    letter = body.get("letter")
    # guess logic here
    save_game(game_id, game)  # save updated state to database
    return web.json_response(game)
```

```python
WORD = "flower"

@routes.post("/guess")
async def post_guess(request: web.Request) -> web.Response:
    body = await request.json()
    letter = body.get("letter")
    guesses = body.get("guesses", [])
    lives = body.get("lives", 5)
    status = body.get("status", ["-"] * 6)

    if letter in guesses:
        return web.json_response({"error": "already guessed"}, status=400)

    guesses.append(letter)
    if letter in WORD:
        for i, l in enumerate(WORD):
            if l == letter:
                status[i] = letter
    else:
        lives -= 1

    return web.json_response({"guesses": guesses, "lives": lives, "status": status})
```

I have 5 servers running my REST API. A client sends a request to server 1, where does the next request from that client need to go?

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A. Server 1

B. Any of the servers

REST Principles

Uniform Interface



mostly for convention and consistency

Stateless

intertwined server design



many servers

Web API

usually a process that will process the results,

client ↑

endpoint that ↙ server returns JSON

↓

can be RESTful

BOTH

use the internet to send/get information

Buse HTTP and client-server model

Website

client → a browser application

server → endpoint that returns HTML + other stuff

Website Requirements

Domain name → public IP address ← (mapped in the DNS server)

At the public IP address
- a process running that acts like a server
  listening at port 80 HTTP Requests
- respond to GET / with index.html

* need public IP address
  need to be running a server

Hosting Your Own Website

your own personal computer doesn't have a public IP address

1. get a computer that is built to be a server
   - set it up to always be on and running
   - a server code

ISP to get public IP address

X as a Service

Pay someone to do the hard parts

trade off ease vs. control

A cloud service manages and runs hardware (software) for you

↳ a buisness model for renting compute resources on demand (AWS)

Infastructure
platform
software

Infrastructure as a Service - IaaS

get a virtual machine that lives some where else physically

- you don't maintain the hardware
- access through ssh

- you do have to set it up

AWS option to rent a VM with a public IP address

Platform as a Service - PaaS

- VM and the environment and scaling

example - a database through AWS

Software as a Service - Saas

Is has a friendly UI already built

Wix

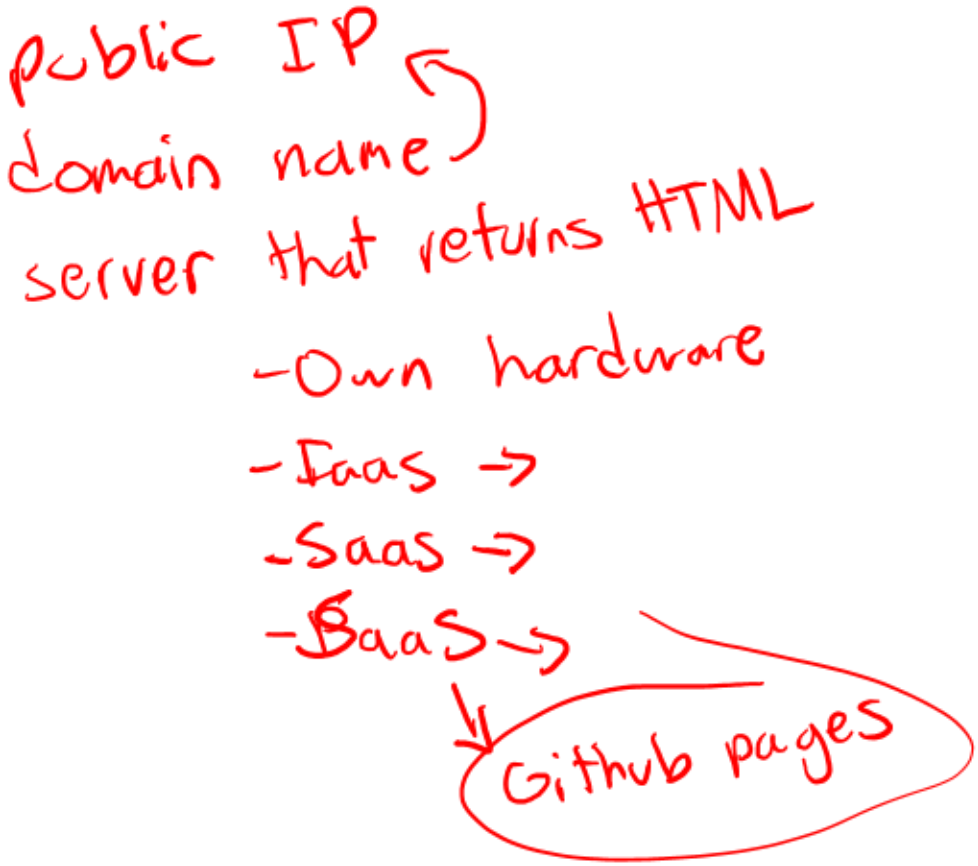
X as a Service Analogy

Iaas - someone gives you a kitchen and oven
- buy ingredients, cook

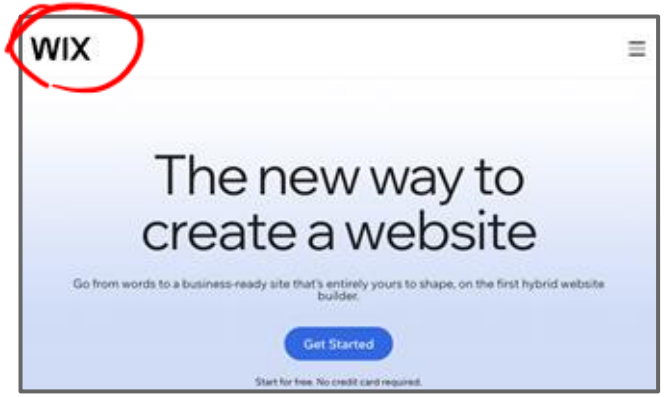
Paas - all of that and premade dough, preheated oven

Saas - order pizza

Hosting your own Website



Hosting a Website - SaaS





- buy a domain name
- host your website

Hosting a Website - PaaS



Vercel

Hosting

get a domain name

website using Next.js

framework

a house with walls and plumbing but you set up the rooms

CS 340 Website

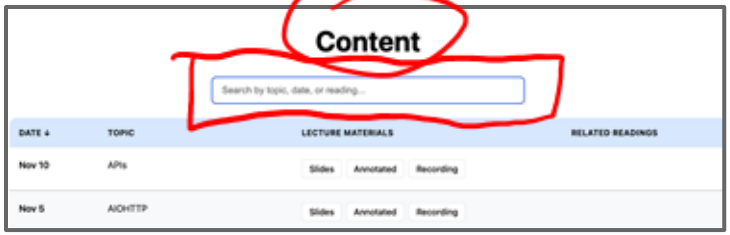
- staff designed the look
- used AI to build it
  - maybe 8 hours







JSON file



I want a website that is just an HTML file, to put it on the internet I can...

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A. Buy a domain name and set it up to grab the file from my personal laptop

B. Use a static hosting site that runs and creates the server for me that responds to GET requests with the html file

C. Get a virtual machine and store the html file on it. Then get a domain name and link it to the virtual machine.

I want to build and deploy a REST API I can...

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A. Get a domain name and then run the server code once on my computer to set it all up with an aiohttp API

B. Get a virtual machine from a cloud provider and connect it with a domain name and then run the aiohttp code on the VM listening on port 80.

C. Use the WIX website to create and deploy the API

How the Cloud Scales Servers

idea 1 → "build hardware that never fails" X

↳ 99.9% good 0.1% fails

- hard drive wear out
- overheating
- someone trips on a cord

↓↓

1 million servers

1000 server fails

"assume hardware fails, design software around it"

Software Solution: Replication

↳ multiple copies across multiple servers

Pros

- if 1 server crashes the data can be recovered
- faster, depending on your location

Cons

- More space — usually 3 copies
- Consistency

Agenda

1. REST APIs

1. X as a service

1. Distributed Computing

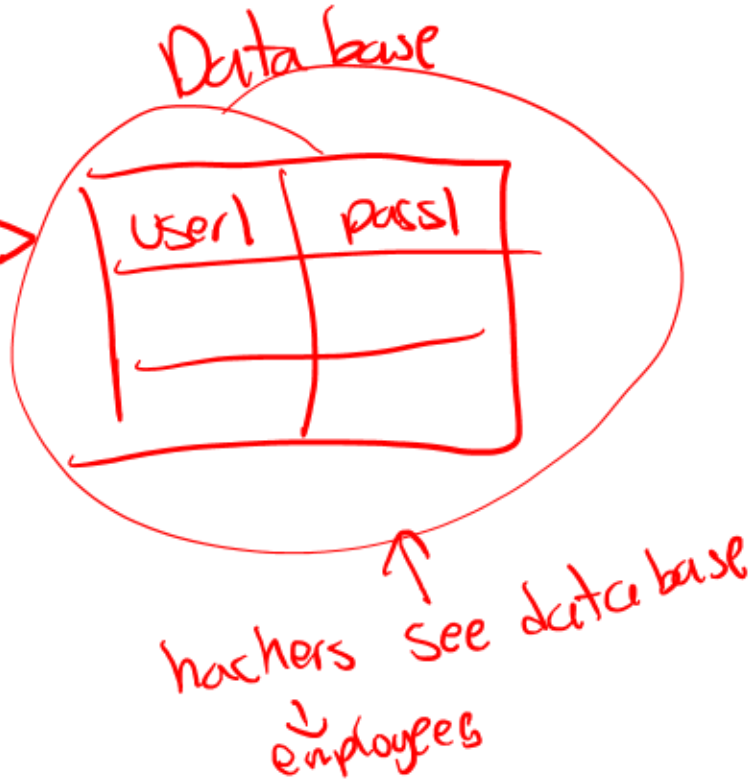
1. Storing Passwords

Storing Passwords

you can create an account

Bad

- for each user

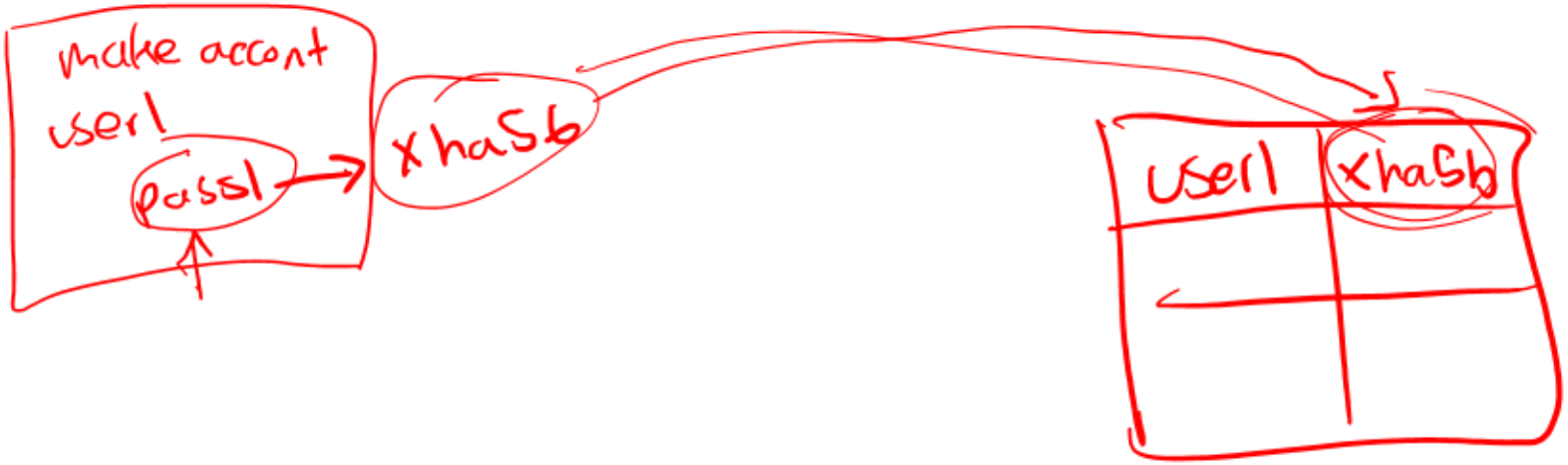


Hash Function

takes in a string and returns a new string.

- always the same new string
- not reversible

Storing Passwords with a Hash



Which of the following is a vulnerability of just using hashing for password storing?

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A. If someone gets access to the database of passwords your password is automatically compromised.

B. If another user has the same password as you and someone figures out that password, your password is compromised.

C. If the hashing function is very slow.

Storing Passwords with a Hash Vulnerability

user A – luther123 → Ax46

user B – luther123 → Ax46

Rainbow table attack

giant text file of common passwords

rockyou.txt

Which best describes a rainbow table attack?

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Q11

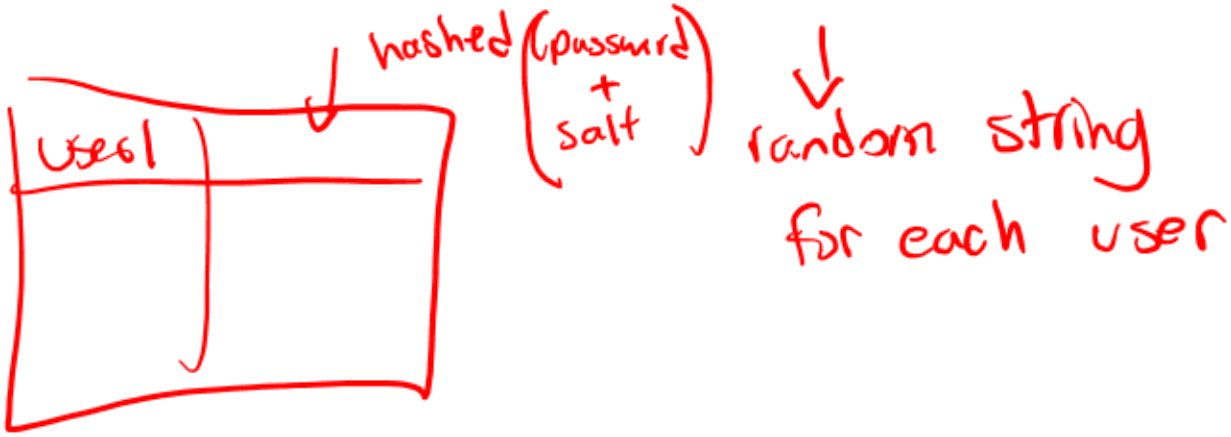
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A. Hashes are pre-computed for popular passwords, then compared to hashes stored for users

A. Passwords are stored in plain text and then the database is compromised

A. Hashes are compared across users to find matches

Storing Passwords with a Salt



What is still a vulnerability even with a hash and salt?

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A. A hacker could guess your password

A. A hacker could see the database with your stored password

A. A hacker could get your password from a different site that doesn't use a hash and salt



Password Tips



7,776

5 words = 90 years

Randomness

Which sequence is least likely to be generated by a truly random number generator?

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A. 1, 2, 3, 4

A. 5, 2, 1, 8

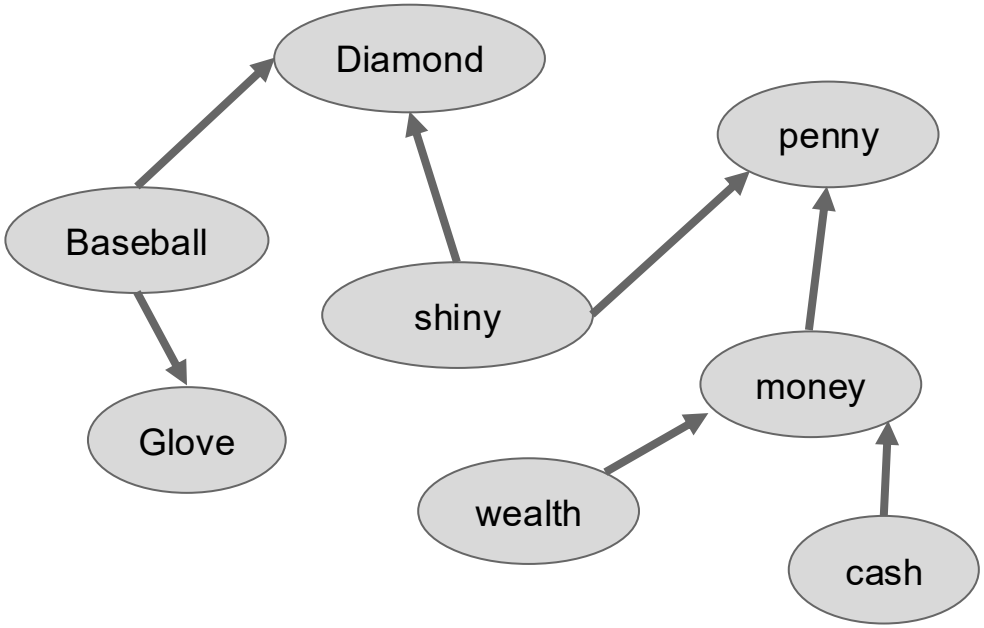
A. 1, 1, 1, 1

A. A or C

A. All the same likelihood

Retrieving a Random Word

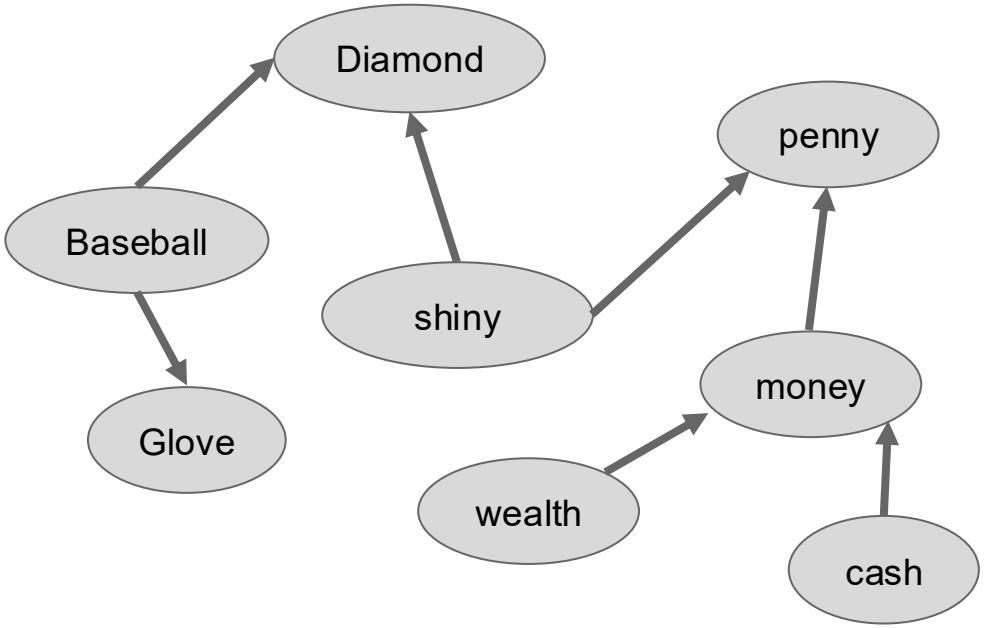
Long Term Memory



Long Term Memory

Topic in Working Memory

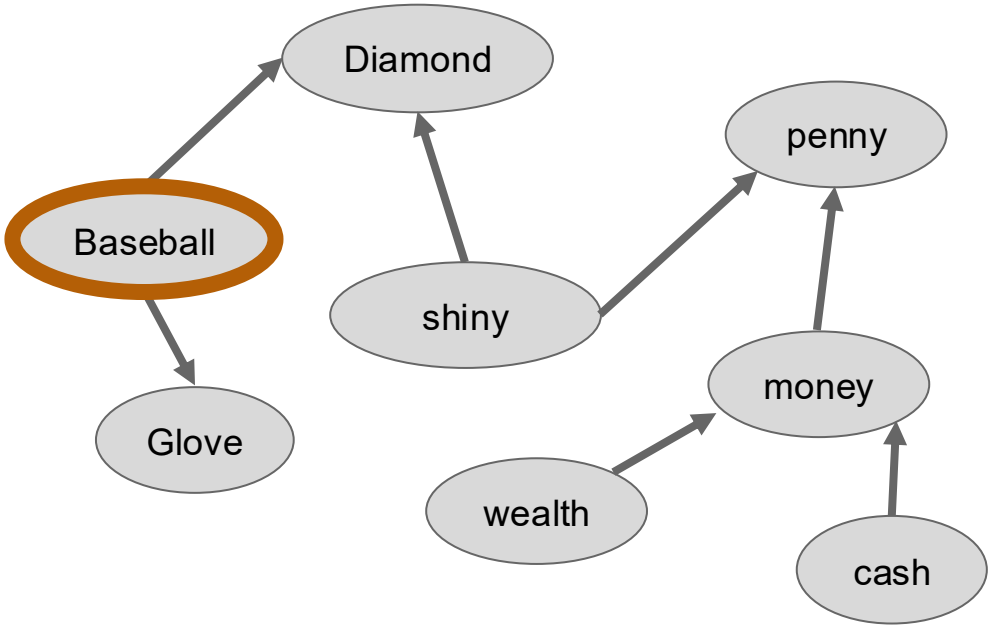
Baseball



Long Term Memory

Topic in Working Memory

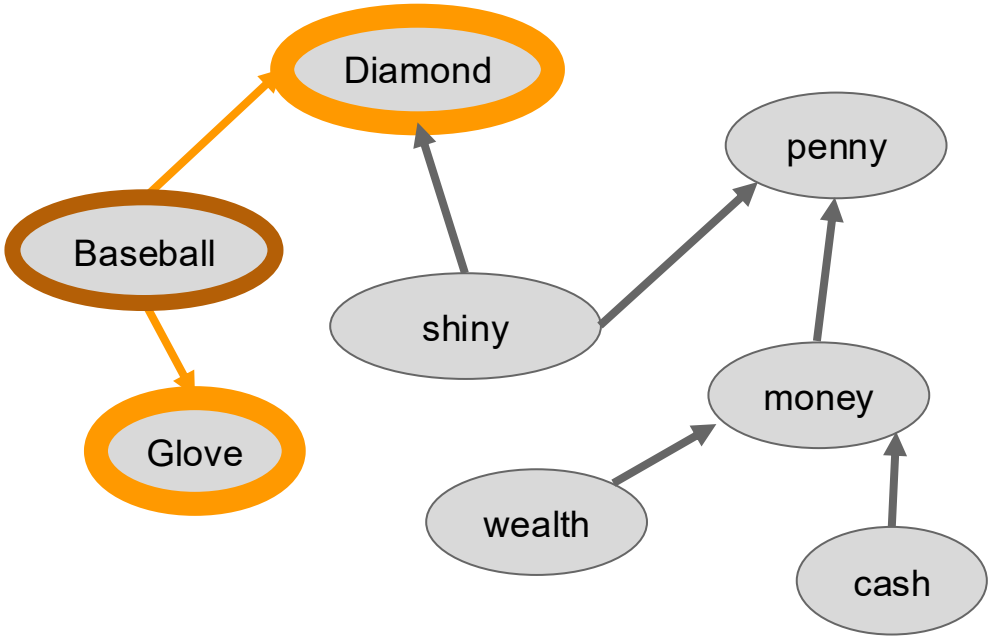
Baseball



Long Term Memory

Topic in Working Memory

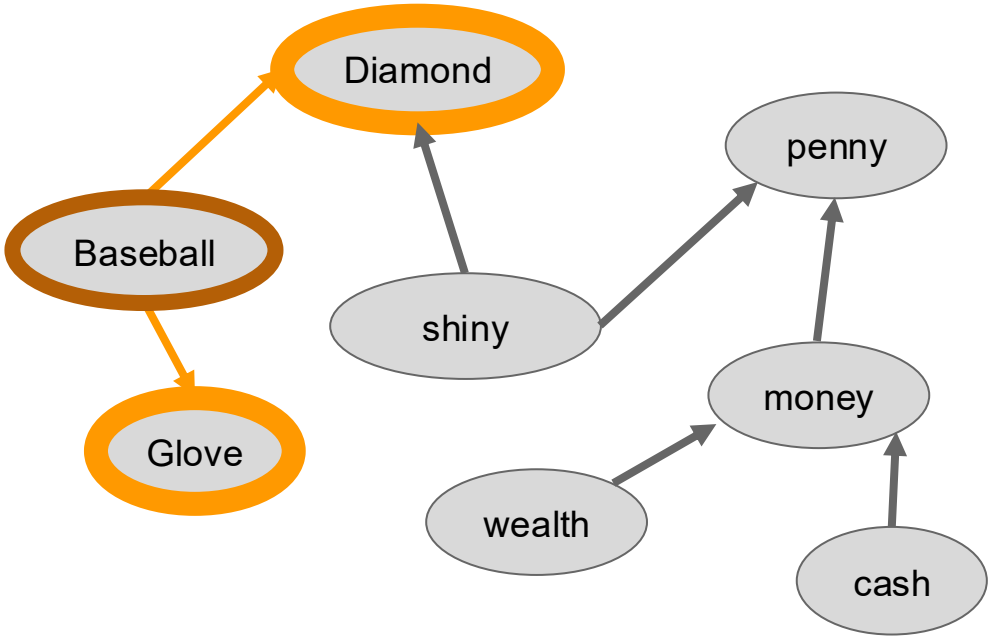
Baseball



Long Term Memory

Request from Working Memory

Give me a random word



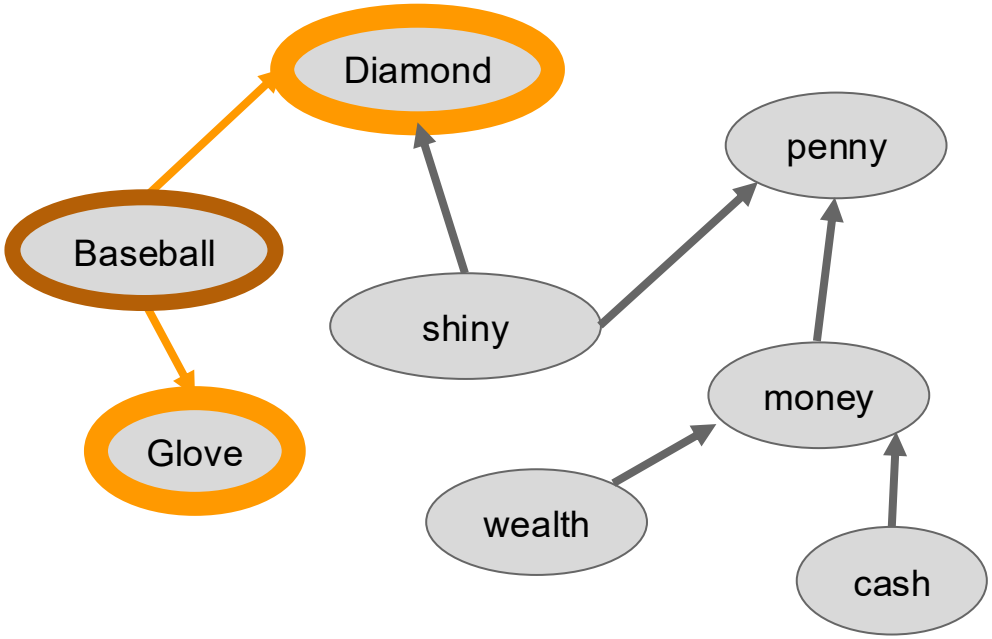
Long Term Memory

Request from Working Memory

Give me a random word

Diamond

Glove



Which idea could generate good random numbers?

A) Analyzing an image of 100 live lava lamps.

A) Ask people on the street for a random word.

A) Atmospheric noise

